

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 9

Topology: Structure Without Measurement

Assignment

Self-Study Curriculum

Consider the “space of all models” for a problem you work on, for instance, the space of all linear regression models with n features (parameterized by $n+1$ real numbers), or the space of all neural networks of a given architecture (parameterized by the weights).

1. What is the dimension of this space?
2. Is it bounded? (Are the parameters constrained?)
3. Is it connected? (Can you continuously deform any model into any other model?)
4. Does it have “holes”, configurations that are missing? If you add regularization, how does the topology change?
5. When you “train” a model by gradient descent, you are tracing a path through this space. What topological properties of the space affect the behavior of the path?

Write this as a structured analysis, not an essay. Diagrams are welcome.